

PERFORMANCE WORK STATEMENT (PWS)

UTILITY MONITORING AND CONTROL SYSTEM (UMCS) SIEMENS GOVERNMENT TECHNOLOGIES DESIGO / APOGEE® BUILDING AUTOMATION SYSTEM (BAS)

10 December 2025

Revision 0

PART 1 GENERAL INFORMATION

1.1 Description of Services. The Contractor shall provide all personnel, equipment tools, materials supervision, and other items and non-personal services necessary to provide the Procurement and Installation (P&I) and/or the Maintenance and Service (M&S) of proprietary Siemens Building Automation Systems Desigo/Apogee® Building Automation System (BAS) including but not limited to: Utility Monitoring and Control Systems (UMCS), Heating, Ventilation and Air Conditioning (HVAC) systems to include chiller/boiler systems installation and/or integration, Supervisory Control and Data Acquisition (SCADA) Systems, Fire Alarm Systems (FAS), life safety systems, other Industrial Control Systems (ICS), automated control systems including chemical / biological / radiological contaminant detection / filtration / response, utilities (electric/ gas/ water/ steam) metering, Electronic Security Systems (ESS), and security and/or force protection measures such as, but not limited to, barriers, fencing, gates, window treatments, hardening, and lighting.

1.2 Background. The U. S. Army Engineering & Support Center, Huntsville (CEHNC) has been designated as the Mandatory Centers of Expertise (MCX) for UMCS and ESS and the Center of Expertise (CX) for HVAC. As a designated MCX and CX, CEHNC provides direct support to Government agencies for the P&I and M&S of UMCS and related systems as described above.

1.3 Objectives. The contractor shall provide technical support, procurement, installation, monitoring, and M&S for UMCS, ESS and related systems. These services include those actions required to preserve and maintain existing systems in such a condition that they may be effectively used for their designated functional purpose.

1.4 Scope. This contract will require the contractor to have the capability to respond to multiple requests simultaneously and to work at diverse locations. The contractor shall provide all engineering and services necessary for the P&I of new and/or upgrades to Direct Digital Control systems, existing HVAC systems, SCADA systems, or other automated control systems on Government facilities. This contract will require monitoring and control and M&S of control systems at Government facilities, municipalities, and educational institutions to meet specific installation requirements. The contract will predominantly be for service efforts, but some tasks may have construction and supply elements.

1.4.1 These services also include Preventive Maintenance (PM), Corrective Maintenance (CM), repair/ replacement/modernization, equipment inventories, condition assessments, identification and/or validation of problems and deficiencies, preparation of work plans describing method of correction/repair, actual execution of work plan to repair failed or failing systems and associated equipment, facility management support, project management, central monitoring services, minor construction work within or exterior to a facility, development of PM plans, development of recommended spare parts lists, and other activities associated with operation, maintenance, and repair. Facility maintenance/repair/renovation shall conform to the requirements of the Unified

Facilities Guide Specifications (UFGS) and Unified Facilities Criteria (UFC) as per the PWS issued for each service requirement and/or other applicable codes and standards.

1.4.2 Services will include, but are not limited to, developing of equipment installation engineering plans, providing hardware and software, providing system cybersecurity engineering and implementation, system testing and check-out, performing facility modification and minor construction required to support equipment installation, providing technical solutions that incorporate applicable force protection measures, interfacing with and installing new data transmission equipment, determining compatibility of new equipment with existing systems and integrating them, preparing system and component documentation, providing operation and maintenance manuals, providing training, warranting systems, maintaining and servicing systems, retro-commissioning systems, operating and monitoring systems, preparing after action and other reports, providing as-built installation drawings, system evaluations, configuring / programming systems, system administration, provide inspections/functional tests / recommendations on control systems, and other technical and programmatic support services.

1.4.3 Period of Performance. The ordering period for this contract is sixty (60) months, comprising of a three (3) year base period and one (1) two (2) year optional periods. The capacity of the SATOC is \$630M. Place of Performance. Contractor is responsible for performing the services required under this contract at the location as specified. These UMCS services are required to support Department of Defense (DoD) installations or other Government facilities. All Continental United States (CONUS locations) as well as Alaska, Hawaii, Germany, Italy, Korea, Japan (including Okinawa) and Africa – Tanzania. Other locations outside the Continental United States where Desigo/Apogee® BAS are previously installed. Other foreign jurisdictions outside the Continental United States may be added to the geographic scope of the base Single Award Base Order Contract (SATOC) if deemed appropriate after an analysis of relevant statutes and regulations, including the Competition in Contracting Act (CICA), and approval of a FAR part 6 J&A if appropriate, and analysis of international agreement terms between the United States and the respective foreign jurisdiction. It is recognized that foreign country contracting involves some unique challenges when preparing solicitation(s) (e.g., Japan requires indirect contracting and Korea requires invited contractor and technical representative status local clauses to be incorporated), so coordination will be done with the Office of Counsel to ensure all appropriate clauses and provisions are included. If the contracting officer, in consultation with Office of Counsel, determines that the specific SOFA or international agreement has a requirement that cannot be met with the contract, then the contract will not be used for that requirement and an alternate acquisition vehicle must be used.

1.5 Wage Rate Requirements (Construction) (WRRC) shall be utilized for construction-based labor efforts. These activities include but are not limited to high/medium voltage wiring, excavation, trenching, removing/repairing walls, flooring, framing, foundation work, masonry, plumbing, lighting replacement, demolition (if other construction work is to follow), painting, replacement of boilers/chillers/air handling units (not including purchase of equipment), installation of new fire alarm systems, and any other work required to be performed by a construction trade. Service Contract Labor Standards (SCLS) wage rates shall be utilized for non-professional service-based labor efforts if the task order is predominantly services. Service activities include but are not limited to the design, project oversight, commissioning, engineering, meter installation, programming, test and balancing, survey, low voltage wiring not requiring an electrician, control system components (motors, actuators, sensors, controls, etc.), repairs to fire alarm systems/chillers/boilers/air handling units, repair/replacement to other ancillary equipment, installation/repair of electronic security systems, and any other work of a services nature that is not required to be performed by a construction trade. SCLS/WRRC Wage Rate Applicability by Location: WRRC wage rates are only applicable within the United States. SCLS rates are applicable within the United States, Northern Mariana

Islands, Guam, Puerto Rico, American Samoa, Johnson Island, the U.S. Virgin Islands, Wake Island, and Outer Continental Shelf Lands.

1.5.1 Quality Control Program (QCP). The contractor shall develop and maintain an effective quality control program to ensure services are performed in accordance with (IAW) this PWS. The contractor shall develop and implement procedures to identify, prevent, and ensure non-recurrence of defective services. The contractor's QCP is how he assures himself that his work complies with the requirements of the contract. At a minimum, the contractor shall develop quality control procedures that address the areas identified in Appendix A, Performance Requirements Summary (PRS).

1.5.1.1 Corporate Quality Control and Surveillance Plan. Contractor shall develop, document, and implement an effective quality control and surveillance program for this contract and document in a plan. The Quality Control QC Plan shall be submitted within 30 calendar days after contract award and shall include at a minimum the following documents: (a) Corporate QC Manager appointment letter, (b) QC organizational structure (organization chart), (c) inspection procedures, (d) recordkeeping procedures, (e) report templates, and (f) any other information as required to describe the QC program. At a minimum, the contractor shall follow an inspection process (e.g., three phase inspection process (preparatory, initial, and follow-up) or other Government approved inspection methodology) and include sample forms to be filled out for each definable feature. After acceptance of the QC Plan, the contractor shall submit any proposed changes to the Government, tracked through Resident Management System (RMS)/Quality Control System (QCS) for approval by the UMCS Program Manager (PgM) or Contracting Officer Representative (COR).

1.5.1.2 Site Specific Quality Control Plan and Surveillance Plan. A task/site specific quality control plan shall be developed/adapted to the services specified and submitted as required. The site-specific QC Plan shall include the following documents at a minimum: (a) The QC Manager appointment letter (include duties and authorities at a minimum), (b) summary of inspection procedures to be followed, (c) sample inspection forms (preparatory, initial, follow-up) or other Government approved inspection methodology forms, and (d) the list of definable features of work that the inspections will be based off. It is the responsibility of the contractor to maintain the inspection reports and to submit them to the Government via email within five (5) calendar days of completion. After acceptance of the QC Plan, the Contractor shall submit any proposed changes to the Government, tracked through Resident Management System (RMS 3.0) for acceptance by the COR.

1.5.2 Quality Assurance (QA). The Government will evaluate the contractor's performance under this contract in accordance with (IAW) with the Quality Assurance Surveillance Plan (QASP) and will include an evaluation of the contractor's performance. This plan is primarily focused on what the Government must do to ensure that the contractor has performed IAW the performance standards. It defines how the performance standards will be applied, the frequency of surveillance, and the minimum acceptable defect rate(s). The Government may conduct QA inspections at any time on all phases and types of work performed. If any Government QA review identifies a process/work product failure or identifies noncompliance with safety practices, the contractor will be issued a Contract Deficiency Report (CDR). Each CDR will be annotated as a critical nonconformance, major nonconformance, or minor nonconformance.

Definitions and required contractor response times follow:

1.5.2.1 Critical Nonconformance: A nonconformance that is likely to result in hazardous or

unsafe conditions for individuals using, maintaining, or depending upon the supplies or services, or is likely to prevent performance of a vital agency mission. Contractor is provided 24 hours to provide written response to the CDR.

1.5.2.2 Major Nonconformance: A nonconformance, other than critical, that is likely to result in failure of the supplies or services, or to materially reduce the usability of the supplies or services for their intended purpose. Contractor is provided three (3) calendar days to provide written response to the CDR.

1.5.2.3 Minor Nonconformance: A nonconformance that is not likely to materially reduce the usability of the supplies or services for their intended purpose or is a departure from established standards having little bearing on the effective use or operation of the supplies or services. Contractor is provided up to ten (10) calendar days to provide written response to the CDR.

1.5.3 Security Requirements. This contract may require the handling and generation of classified information. Security requirements appropriate for each service will be delineated. Some service locations may require a security level of SECRET, and some may require a higher level than SECRET. The contractor shall meet the Industrial Security requirements of the National Industrial Security Program Operating Manual (NISPOM) for Safeguarding Classified Information (DoD 5220.22M). The clearance level will be determined by the User Agency. A DD Form 254, Contract Security Classification Specification, shall be provided. No classified work is authorized without an approved DD 254 and the appropriate personnel and facility security clearance. Personnel security clearances will be obtained by the contractor IAW the NISPOM. Facility security clearances will be obtained by the User Agency Security Office.

1.5.3.1 Information Security. Contractor shall provide adequate security for all unclassified DoD information passing through non-DoD information systems including all subcontractor information systems utilized on contract. Unclassified DOD information shall only be disseminated within the scope of assigned duties and with a clear expectation that confidentiality will be preserved. Examples of such information include the following: non-public information provided to the Contractor, information developed during the contract, and privileged contract information (for ex: program schedules, contract-related tracking). Below are safeguards the contractor, and all subcontractors, shall abide by to protect Government information.

- a) Do not process DoD information on public computers (e.g., those available for use by the public in kiosks or hotel business centers) or computers that do not have access control.
- b) Protect information by at least one physical or electronic barrier (e.g., locked room, login, and password) when not under direct individual control.
- c) Sanitize media (e.g., overwrite) before external release or disposal.
- d) Encrypt all information that has been identified as controlled unclassified information when it is stored on mobile computing devices such as laptops and personal digital assistants, or removable storage media such as thumb drives and compact disks, using the best available encryption technology.
- e) Limit information transfer to subcontractors or teaming partners with a need to know and commitment to at least the same level of protection.
- f) Transmit email and similar communications using technology and processes that provide the best level of privacy available, given facilities, conditions, and environment.
- g) Do not post DoD information to Web site pages that are publicly available or have access limited only by domain or Internet protocol restriction.
- h) Provide protection against computer network intrusions and data exfiltration, minimally including the following: updated malware protection devices, prompt application of security-relevant software patches/service packs/hot fixes, and monitoring and control of inbound and

outbound network traffic as appropriate.

i) Report loss or unauthorized disclosure of information IAW contract or agreement requirements and mechanisms.

j) The contractor shall not release any information to the public without prior approval by the CEHNC Office of Public Affairs (i.e., trade publications and marketing information).

1.5.3.2 Site Specific Security Requirements. When contractor personnel are at any Government site, installation, or facility, they shall follow all security procedures and processes of the installation as stated and directed by the Government personnel at the site. The contractor's security requirements for work in foreign countries will be based on the requirements set by the combatant commander of the Area of Responsibility in which the work will be performed. The items that will be covered will include but are not limited to: Transportation into and out of the operating area, required clothing and equipment, transportation security while moving in the operating area, specific employee training prior to entering the operating area, etc. All security requirements will be further detailed at the time of TO awards.

1.5.3.3 Information Technology Systems. For services requiring access to Government Information Technology (IT) Systems, but not to classified information, the following requirements apply:

1.5.3.4.1 All Contractor employees who are U.S. citizens working under this contract who require access to IT systems (stand-alone computers, network computers/systems, e-mail, etc.) shall, at a minimum, be designated into an IT level I, II, or III position IAW Army Regulation (AR) 25-2, Information Assurance. The investigative requirement for an IT level I position is a favorable Single Scope Background Investigation; IT level II and IT level III positions require a favorable National Agency Check with Law and Credit.

1.5.3.4.2 A Common Access Card (CAC) is also required for access to Government IT systems. Proof of the appropriate investigative requirement/background check will be verified through the Joint Personnel Adjudication System by the CEHNC security personnel.

1.5.3.4.3 Non- United States. Citizens. Foreign nationals who work on a Corps of Engineers' contract in the United States (U.S.) shall be approved by the Headquarters U.S. Army Corps of Engineers (HQUSACE) Foreign Disclosure Officer before beginning work on the contract. This approval requirement also includes subcontractor employees. The Contractor shall submit to the KO the names of all foreign nationals proposed for performance under this contract along with documentation to verify that he or she is legally admitted into the U.S. and has authority to work. This information will be forwarded to the CEHNC Security Officer for approval.

1.5.4 In accordance with FAR 52.236-13, Accident Prevention, the Contractor shall comply with the latest version of U.S. Army Corps of Engineers (USACE) Safety and Health Requirements Engineering Manual (EM) 385-1-1 (EM 385-1-1), including any interim revisions in effect at the time of the solicitation, as well as any current state, federal, and local regulations. EM 385-1-1 and its changes are available at: <http://www.publications.usace.army.mil/USACE-Publications/Engineer-Manuals/>

1.5.4.1 Corporate Safety and Health Program. The Contractor is required by regulation to develop and maintain a written Corporate Safety and Health Program (SHP) in compliance with the requirements of the Occupational Safety and Health Administration (OSHA), EM 385-1-1, 29 CFR 1910, 29 CFR 1926, 29 CFR 1960, and FAR 52.236-13.

1.5.4.2 Site Visit Abbreviated Accident Prevention Plan (SVAAPP). The Contractor will be required to submit an SVAAPP prior to commencement of any site visit which are not covered

under an Accident Prevention Plan. The Contractor shall prepare a “fill-in-the-blank” template SVAAPP to be used for all site visits. The SVAAPP shall be adapted to sites specified in the individual task/delivery orders and shall be prepared prior to each site visit. SVAAPP’s shall be submitted electronically to the CEHNC Safety Office and Project Manager (Adobe PDF format) for review and acceptance. Upon acceptance, by the CEHNC Safety Office, the Contractor shall submit final SVAAPP via e-mail, including tracking submittal via Form 4025 through QCS/RMS. Final acceptance threshold is an approved transmittal of the SVAAPP via QCS/RMS (note: an e-mail from the CEHNC Safety Office of approval does not constitute full acceptance of an SVAAPP) - in accordance with each task order. The SVAAPP shall be submitted to the CEHNC Safety Office for acceptance before any work under this contract is awarded. An example of an SVAAPP can be obtained by contacting the CEHNC Safety Office. The SVAAPP shall be approved prior to the Contractor performing any surveys on-site. Submission or approval requirements may be updated at the TO level.

1.5.4.3 Site Specific - Accident Prevention Plan (APP). The Contractor shall submit an APP prior to any installation/maintenance activities beginning at the site. APPs shall address applicable items listed in Appendix A (Minimum Basic Outline for Accident Prevention Plans) of EM 385-1-1 in addition to the other applicable requirements of EM 385-1-1 and OSHA requirements. The plan shall be site specific and shall also address any unusual or unique aspects of the project or activity for which it is written. The Contractor shall coordinate with the facility representatives or facility contractor, on first aid, emergency phone numbers & response plans, site evacuation plans, lock out tag out procedures, any permit required procedures, obtaining a list of previously identified confined spaces, evacuation drills, hazardous/toxic environments/agents etc. This information shall be included in the APP. The APP shall also include site specific Activity Hazard Analysis (AHAs) for the project. The APP is the document that details how the safety and health requirements of OSHA; other regulatory codes and documents; and EM 385-1-1 are to be accomplished. Accident Prevention Plans (APP, IAW EM 385-1-1, Appendix A) shall be submitted in accordance with each task order. The APP shall be approved prior to the Contractor mobilizing on-site to perform work. Submission or approval requirements may be updated at the TO level.

1.5.4.4 Addendums to the APP - The Contractor is responsible to verify that any Proposals, Service Orders, Change Orders, and Contract Modifications are covered under the current accepted APP. If the pending work effort is NOT covered under the initial task order APP, the contractor shall submit, for Government review and acceptance, an addendum to the APP.

1.5.4.5 Activity Hazard Analysis (AHA) - Before beginning each activity, task or Definable Feature of Work (DFOW) involving a type of work presenting hazards not experienced in previous project operations, or where a new work crew or subcontractor is to perform the work, the Contractor(s) performing that work activity must prepare an AHA. AHAs must be developed by the Prime Contractor. If the Prime Contractor requires a subcontractor or any sub tier- contractor or supplier performing the work, to develop an AHA, it shall be provided to the Prime Contractor SSHO for review and approval before submitting to the Huntsville Center Safety Office for review and acceptance. The Prime Contractor shall review the AHA list periodically (at least annually in conjunction with an APP review) and update as necessary when procedures, scheduling, or hazards change.

1.5.4.6 Site and Safety Health Officer (SSHO). The Contractor must employ a minimum of one Competent Person at each project site to function as the Site Safety and Health Officer (SSHO). The SSHO does not have to be a direct hire employee of the prime Contractor. Alternate SSHOs may be approved at the discretion of the CEHNC Safety Office and are not required to

be a direct hire employee of the prime. Contractor shall declare on the APP and demonstrate, through training documentation maintained at the site, that the SSHO meets the requirements of EM 385-1-1,

01.A.17. The SSHO is required to always be onsite during the performance of work. The duties of the SSHO may be fulfilled as an additional job site responsibility (dual hat) by another Competent Person who is not the Supervisor; however, these collateral duties shall not overcome, inhibit or restrict the effective administration of the Safety and Health Program. Moreover, the functions and roles of the SSHO shall be a singular, full-time responsibility (no other collateral duties) during high hazard activities to include, but not limited to: crane operations, excavations greater than 5-feet, confined space entry, and electrical work to include Lock-out-Tag-out (LOTO) in accordance with the Activity Hazard Analysis (AHA) as prepared per the accepted APP. The Contractor must demonstrate that this employee's non-safety duties are transferred to another employee when full-time SSHO responsibilities are warranted per the risk activities. The SSHO must be at the location of the high hazard activities while they are being conducted.

1.5.5 Exposure Hour and Mishap Reporting. Exposure Hour Reporting (Monthly Man-hour reporting) - In accordance with EM 385-1- 1, the Contractor shall complete a monthly summary report of accident experience, exposure, Restricted Duty (RD) and Lost Workdays (LWD). The report shall be on the electronic data collection tool provided by the CEHNC Safety described as total hours associated where workers are exposed to the activity hazards and forwarded to the Government Designated Authority (GDA) no later than close of business (COB) the 7th day of the following month.

a) The Contractor shall submit a monthly report to the GDA, no later than close of business (COB) the 7th day of the following month, that includes a summary by month and year to date of exposure hours per base contract number and per program. (i.e. UMCS, Metering, and ESS shall have separate monthly summary reports). Government may change the form or submission method as necessary.

b) The Contractor shall submit a monthly exposure hour report in RMS/QCS at the TO level no later than the COB the 7th day of the following month, if required at the TO level.

The information cut-off date will be the last day of each month. The monthly submission must include the title of the report, i.e., Accident Exposure Data Report, contract number, TO number, project site, month and year for which the report is made, a point of contact listing both email address and telephone number, and number of lost workday accidents to include total days lost. If no hours are worked on the project/task, a report showing "zero (0)" is required.

1.5.5.1 Mishap Reporting – Mishaps shall be reported in accordance with the EM 385-1-1 paragraph 01.D. The Contracting Officer and Huntsville Center Safety Office shall be notified as soon as practical, but no more than twenty-four hours, after any mishaps, including recordable accidents, incidents, and near misses. All mishap notifications will be submitted in accordance with the task order.

1.5.5.2 Preparatory and Initial Control Phase Meetings (Kick-off Meetings) – Per the EM-385-1-1 Section 01.A.14.c & Appendix A; Understanding that the APP is submitted prior to work starting and that some project-specific information is not yet known (e.g., subcontractors to be used, type of fall protection to be used at a certain point in time, specific cranes or other load handling equipment (LHE) to be brought on site, etc.), the known information shall be provided and additional information added at each preparatory phase/meeting along with the appropriate

Activity Hazard Analyses (AHAs). Work shall not begin until the AHA with RAC for the work activity has been accepted by the GDA and discussed with all engaged in the activity, including the Contractor, subcontractor(s), and Government on-site representatives at preparatory and initial control phase meetings. Notification of these meetings shall be extended to the USACE PM, KO, Engineers and Safety Representative for the given task order.

1.5.5.3 Traffic Control Plan. For any work that is to take place in or near roadways the contractor shall submit a traffic control plan for approval prior to commencing any work. This traffic control plan shall be developed IAW EM 385-1-1, state specific guidelines and requirements and installation specific requirements.

1.5.6 Post Award Conference/Periodic Progress Meetings. The contractor agrees to attend any post award conference convened by the contracting activity or contract administration office IAW FAR Subpart 42.5. The KO, COR, and other Government personnel, as appropriate, may meet periodically with the contractor to review the contractor's performance. At these meetings the Government will apprise the contractor of how the Government views the contractor's performance and the contractor will apprise the Government of problems, if any, being experienced. Appropriate action shall be taken to resolve outstanding issues. These meetings shall be at no additional cost to the Government.

1.5.7 Program Manager (PgM). The contractor shall provide a PgM who shall be responsible for the performance of the work. The name of this person and an alternate who shall act for the contractor when the manager is absent shall be designated in writing to the KO. The PgM or alternate shall have full authority to act for the contractor on all contract matters relating to daily operation of this contract.

1.5.8 Identification of Contractor Employees. All contract personnel attending meetings and working in other situations where their contractor status is not obvious to third parties are required to identify themselves as such to avoid creating an impression in the minds of members of the public that they are Government officials. They must also ensure that all documents or reports produced by contractors are suitably marked as contractor products or that contractor participation is appropriately disclosed.

1.5.9 Travel

1.5.9.1 Locations: In support of various tasks and as specified, the contractor shall be required to travel to DoD facilities located worldwide. Travel shall be performed IAW FAR 31.205-46.

1.5.9.2 Travel Outside the Continental United States Immunization Requirements: As specified in each TO, the contractor shall be required to travel to locations both shore and afloat. Contractor employees who deploy to locations that require immunizations shall do so IAW DoD Foreign Clearance Guide or other agreements that the U.S. may have with the country.

1.5.9.3 Letter of Authorization. Some travel shall require a Letter of Authorization (LOA). As noted in Defense Federal Acquisition Regulation Supplement (DFARS) 225.371-3(e), a LOA is necessary to enable a contractor employee to process through a deployment processing center; to travel to, from, and within a theater of operations; and to identify any additional authorizations and privileges. As required, the contractor shall initiate a LOA for each prospective traveler. If required, the contractor shall use the Synchronized Pre-deployment & Operational Tracker (SPOT) web-based system, at <http://www.dod.mil/bta/products/spot.html>, to enter and maintain data with respect to traveling/deployed personnel, and to generate LOAs.

If it is in the Government's interest, the contractor may also initiate a LOA request to provide an official traveler access to Government facilities and to take advantage of travel discount rates IAW Government contracts and/or agreements. All privileges, services, and travel rate discount access are subject to availability and vendor acceptance. LOAs shall be signed/approved by the KO who is registered in the SPOT.

1.6 Base Contract Submittals. The contractor shall submit documents in the following format in the table 1 below and will be tracked through RMS (3.0) within thirty days of base contract award to the UMCS Program Manager. The documents shall be submitted by task order number format W912DY21FXXXX, minimum obligation. After approval, all proposed changes by the contractor shall be submitted under the task order number. W912DY21FXXXX.

Table 1

Submittals	Tracked in RMS (3.0)	Form 4025 Required	CD/DVD	E-mail	
				PgM	Safety Office
Corporate Safety and Health Program	Yes	Yes	√	√	√
Corporate Quality Control and Surveillance Plan	Yes	Yes	√		
Corporate Management Plan	Yes	Yes	√		
Unless otherwise specified, all report formats shall be Microsoft Office (Word and Excel) and /or searchable Adobe PDF					

PART 2 DEFINITIONS & ACRONYMS

A&A – Assess and Authorize
 AAPP – Abbreviated Accident Prevention Plan
 ACA – Agent for the Certification Authority
 ACE-IT – Army Corps of Engineers Information Technology
 After Hours – Hours outside of Normal Working Hours
 AHA – Activity Hazard Analysis
 AO – Authorizing Official
 AODR – Authorizing Official Designated Representative
 APMS – Army Portfolio Management System
 APP – Accident Prevention Plan
 AR – Army Regulation
 ATC – Authority to Connect
 ATO – Authority to Operate
 CAC – Common Access Card
 CAT - Category
 CCI – Control Correlation Identifier
 CDRL – Contract Data Requirements List
 CDR – Contract Deficiency Report
 CIA – Confidentiality, Integrity and Availability
 CM – Corrective Maintenance
 CND – Computer Network Defense
 COB – Close of Business
 COR – Contracting Officer Representative
 CPARS – Contractor Performance Assessment Reporting System

CS – Control Systems
CSET – Cyber Security Evaluation Tool
CEHNC – U.S. Army Corps of Engineers, Engineering and Support Center, Huntsville
DA – Department of the Army
DFARS – Defense Federal Acquisition Regulation Supplement
DHS – Department of Homeland Security
DISA – Defense Information Systems Assurance
DoD – Department of Defense
DoDD – Department of Defense Directives
DoDI – Department of Defense Instructions
EM – Engineering Manual
eMASS – Enterprise Mission Assurance Support Service
ER – Engineering Regulation
ESS – Electronic Security Systems
FAR – Federal Acquisition Regulation
FAS - Fire Alarm System
FSC –Federal Service Code
GDA – Government Designated Authority
HVAC – Heating, Ventilation, and Air Conditioning
HQUSACE – Headquarters, U.S. Army Corps of Engineers
IA – Information Assurance
IAM – Information Assurance Manager
IASAE – Information Assurance System Architect and Engineer
IAT – Information Assurance Technical
IAVA – Information Assurance Vulnerability Alerts
IAW – In Accordance With
ID/IQ – Indefinite Delivery / Indefinite Quantity
ICS – Industrial Control Systems
IO – Information Owner
IS – Information System
ISO – Information System Owner
ISSM – Information System Security Manager
ISSO – Information System Security Officer
IT – Information Technology
KO – Contracting Officer
LOA – Letter of Authorization
LWD – Lost Workday
M&S – Maintenance and Service
NEC – Network Enterprise Center
NISPOM – National Industrial Security Program Operating Manual
NIST – National Institute of Standards and Technology
O&M – Operation and Maintenance
OEM – Original Equipment Manufacturer
OO – Ordering Officer
OSHA - Occupational Safety and Health Administration

P&I – Procurement and Installation
PAC – Package Approval Chain
PE – Professional Engineer
PIT – Platform IT
PGI – Procedures, Guidance, and Information
PgM – Program Manager
PM – Preventive Maintenance
POA&M – Plan of Action and Milestones
PPS – Ports, Protocols, and Services
PRS – Performance Requirement Summary
PVT – Performance Verification Test
PWS – Performance Work Statement
QA - Quality Assurance
QASP – Quality Assurance Surveillance Plan
QC - Quality Control
QCP – Quality Control Program
QCS – Quality Control System
RD - Restricted Duty
RFP – Request for Proposal
RMF – Risk Management Framework
RMS – Resident Management System
SCA-A – Security Control Assessor/Army
SCA-R – Security Control Assessor Representative
SCA-V – Security Control Assessor/Validator
SCADA – Supervisory Control and Data Acquisition
Systems SO – Service Order
SPOT – Synchronized Pre-Deployment & Operational Tracker
SSHO – Site Safety Health Officer
SSE – Security System Engineer
SSP – System security plans
TO – Task Order
UFC – Unified Facilities Criteria
UFGS – Unified Facilities Guide Specification
UIC – Unit Identification Code
UMCS - Utility Monitoring and Control System
UR – User Representative
USACE – United States Army Corps of Engineers

PART 3 GOVERNMENT PROPERTY AND SERVICES

3.1 General: Typically, no Government property will be provided

3.2 Publications and Forms. The Government will provide the contractor with all necessary Government publications to include regulations, engineering pamphlets, technical manuals, standards, user's manuals, and forms as necessary for the performance of tasks related to the work described.

PART 4 CONTRACTOR FURNISHED ITEMS AND SERVICES

4.1 General: Except for those items specifically stated to be Government furnished, the contractor shall furnish all labor, supervision, travel, equipment, and materials as required to perform the work.

4.2 Equipment installed, associated manuals, and site documentation/as built drawings: These items acquired or developed by the contractor in the performance of the work under everyone TO shall become Government Property as title vests to the Government upon receipt of payment.

4.3 Security Clearances: Appropriate security requirements will be delineated for each service area. The contractor shall have or be capable of obtaining a facility clearance through SECRET. All key personnel designated shall have or be able to obtain a security clearance of SECRET. Some work and issues may require a higher level of clearance than SECRET. The contractor shall comply with the security requirements of the DoD Contract Security Classification Specification, DD Form 254.

PART 5 SPECIFIC TASKS

5.1 Basic Services. The following requirements and deliverables will be required under the contract. Additional requirements described for each service may complement, emphasize, or add to any requirements or processes mentioned in this paragraph. The Government reserves the right to expand the requirements so long as the requirement does not force the contractor to perform outside the scope of what would be considered reasonably within the bounds of this contract.

5.1.1 Contractor shall participate in a post-award orientation conference at the site prior to installation work. The work will be associated with P&I of and/or the M&S of ICS including but not limited to: UMCS, HVAC systems installation and/or integration, SCADA systems, FAS, life safety systems, other automated control systems including chemical / biological / radiological contaminant detection / filtration / response, utilities (electric/ gas/ water/ steam) metering, ESS, and security and/or force protection measures such as, but not limited to, barriers, fencing, gates, window treatments, hardening, and lighting. HVAC systems may include central plant equipment and systems (e.g., chillers, boilers, cooling towers), distribution systems (e.g., duct work, chilled water piping, hot water piping, drain piping), and unitary heating and cooling equipment (e.g., unitary heat pumps, unitary air conditioners). The work may include design of systems, such as defined above, and will be further clarified at time of service.

5.2 Specific task areas or functions to be performed are listed below. All tasks are to be performed IAW criteria, specifications, drawings, original manufacturer's published procedures (if applicable), and Contract Data Requirements List (CDRL), Form DD 1423 As applicable to specific tasks, the contractor shall:

5.2.1 Participate in a post-award orientation meeting at the site prior to commencing work. Provide a QCP, a Project Schedule and an APP.

5.2.2 Perform an existing site conditions survey to validate the quantities of system equipment and verify existing site conditions. Provide a written report noting all identified discrepancies.

5.2.3 Perform engineering surveys or assessments to develop or verify design drawings and equipment lists. Provide a report identifying equipment incompatibility concerns between the existing and new systems.

5.2.4 Prepare and provide mechanical equipment installation engineering plans.

5.2.5 Prepare and provide electronic equipment installation engineering drawings to include drawings related to installation and assembly of the computers, field equipment panels, data transmission equipment, interfaces, instrumentation, and all other equipment.

5.2.6 Prepare and provide system installation engineering drawings, instrumentation installation drawings, wiring diagrams, procedures for field installation of all equipment, and system testing procedures, as defined in UFC 3-470-01 and UFC 3-410-02, or latest Corps of Engineers technical guidance.

5.2.7 Provide Mechanical Equipment. Install equipment IAW original manufacturer's published procedures and documentation and as required by the specifications.

5.2.8 Provide Hardware and Software. Install hardware and software IAW original software developer's published procedures and documentation and as required by the specifications. Resolve compatibility issues and establish an interface between existing and new systems. All software and software updates shall be approved by the Government prior to installation. Provide graphics for the system IAW criteria and specifications using data from drawings, engineering surveys and manufacturer's data.

5.2.9 Perform factory testing, Performance Verification Testing (PVT), endurance testing, and commissioning IAW the applicable and referenced sections of criteria and specifications in this TO. For PVT and at his/her sole discretion, the KO may require consideration from the contractor equivalent to reimbursing the cost of the Project Engineer and/or other Government PVT witness (including, but not limited to the labor and travel of personnel conducting the subsequent PVT reassessment) in the form of a credit to the contract price if the contractor does not pass the initial PVT attempt.

5.2.10 Perform minor ancillary work required to support equipment installation.

5.2.11 Provide all data transmission equipment IAW specifications, drawings, and manufacturer's published documentation. Install in compliance with Network Enterprise Center (NEC) or its equivalent authority, Department of Defense (DoD) and Component Specific applicable certification and accreditation requirements, policies, and procedures.

5.2.12 Interface to all existing data transmission equipment IAW specifications, drawings, and manufacturer's published documentation. Comply with all requirements of the DoD, Component, NEC, or its equivalent authority related to interfacing with the existing communication system. Provide technical data, equipment, and software in compliance with Cybersecurity policy and requirements.

5.2.13 Provide all required Operation and Maintenance (O&M) Manuals.

5.2.14 Provide training as required by the applicable sections of the specifications listed in Section H, Exhibit F of the RFP.

5.2.15 Provide a System Warranty. Provide a one-year unconditional, excluding acts of God and negligence of Government personnel, parts and labor warranty from time of Government acceptance (i.e. successful PVT) for the new components of the entire system: Mechanical equipment, ancillary components, subsystems, data transmission system, and all contractor furnished software.

5.2.16 Provide after-action reports, accident reports, minutes of each meeting, confirmation notices of communications, field inspection reports, calibration reports, and installation progress reports. Copies shall be furnished to the Government within five working days or sooner, if required, after each specific action.

5.2.17 Provide final As-Built Drawings. All drawings shall be generated in electronic media compatible with the latest version of MicroStation, AutoCAD, or as defined in individual TOs. Drawings shall be delivered within 30 calendar days after the written acceptance of the system by the Government.

5.2.18 Provide software licensing and configuration tools related to UMCS, ESS, SCADA, other control and monitoring systems and HVAC systems as further clarified by individual TO.

5.2.19 Perform special engineering services and studies related to UMCS, ESS, SCADA, other control and monitoring systems and HVAC systems as further clarified by individual TO.

5.2.20 Provide all M&S and/or operation of systems as specified.

5.2.21 Comply with Cybersecurity Requirements. Prepare and provide system cybersecurity configuration specifications as defined in UFC 4-010-06 Cybersecurity for Facility-Related Control Systems, with Change 1 and UFGS 25-05-11 Cybersecurity for Facility-Related Control System, and / or latest Corps of Engineers technical guidance or another authoritative DoD and / or Federal agency. Provide system, security, and network engineering services, RMF (Assess Only or Assess and Authorize) process services, project specifications, and deliverables to support system owner required cybersecurity activities to obtain operational and/or connection approvals. Cybersecurity requirements will be further clarified by individual Task Order level Performance Work Statements. Refer to Appendix B for additional clarification on task order level requirements.

5.3 Individual Requirements

5.3.1 The contractor shall perform the work under individual requirements. Each individual requirement will identify the specific work to be performed. Individual requirements may add additional technical requirements as necessary to define the work, including Specifications, Technical Manuals, Engineer Instructions, and EMs. The performance period will be negotiated and determined for each individual requirement. Contractor shall prepare and submit a detailed project schedule in Primavera (unless otherwise specified) that shows an appropriate breakout of

activities and includes any key milestones that may be stated in individual requirements.

5.4 Information Support and Documentation.

5.4.1 All required technical information and data shall be delivered IAW the requirements, quantities, and schedules set forth in the contract, CDRL DD Form 1423, Data Item Description DD Form 1664, or as otherwise specified or required. The requirements for contents and detail of operation and maintenance manuals, testing documentation (including level and detail) shall be as described in applicable UFGS and shall be further clarified in the individual TO PWS.

5.4.2 Contractor shall be responsible for preparing after-action and other reports to include progress reports, minutes of all meetings, confirmation notice for telephone calls, field inspection reports, calibration reports and installation reports. Copies of all such documents shall be furnished, within five working days, to the points of contact identified in the TO after each specific action. Project status shall be formally reported monthly.

5.5 Continuation and Completion of Work.

5.5.1 The contractor shall conduct all work under the direction of a COR appointed by the KO. This representative shall be responsible for the coordination of all work under each individual TO. The contractor's company officer or representative designated as PgM shall be responsible for the complete coordination of all work under this contract. All work shall be accomplished IAW the contractor's approved QC Plan. The Government may elect to provide third party technical inspection.

5.5.2 Warranty: The systems, all its components, subsystems, data transmission system, and all contractor-furnished software shall be covered by warranty. The contractor shall warrant all products and services provided under each TO, material and labor, for a period of one year from the date of Government acceptance (i.e. successful PVT), or the offeror's standard commercial warranty period, whichever is greater. The contractor shall provide the necessary support required to maintain the warranty. Upon any product/service failure during the warranty period, contractor's response time to restore system service shall be within the timeframes indicated herein. "Mission Critical Outages" will require a maximum of 4-hour response time during normal working hours and maximum of 16-hour response time for after hours (defined as outside "normal" working hours - 7:00 am through 4:00 pm) and one day to restore system operation. All other outages will require a maximum of three (3) working days response time and seven working days to restore system operation. System outages assessed as "Mission Critical" will be at the discretion of the KO or COR.

5.5.3 Contractor's warranty letter shall be submitted after successful completion of PVT. Warranty letter shall indicate at a minimum: Standard project warranty dates and points of contact, equipment warranties and points of contact, list of equipment (make/model/serial number of equipment) with Original Equipment Manufacturer (OEM) warranties that extend beyond the project warranty period, and OEM warranty documentation.

5.6 Resident Management System (RMS). The Government will use RMS for Windows to assist in its monitoring and administration of requirements. The Government will use the RMS to assist in its monitoring and administration of TOs. The contractor uses the Government-furnished Construction Contractor Mode of RMS, referred to as RMS QCS, to record, maintain,

and submit various information throughout the contract period. The contractor mode user manuals, updates, and training information can be downloaded from the RMS web site (<https://rms.usace.army.mil>). The joint Government-Contractor use of RMS facilitates electronic exchange of information and overall management of requirements. QCS provides the means for the contractor to input, track, and electronically share information with the Government in the following areas: Administration, Finances, Quality Control, Submittal Monitoring, Scheduling, and Import/Export of Data.

5.7 Performance Measures.

5.7.1 Performance Measures for each requirement will be agreed upon before award depending on the nature of services and/or work required. The Government will monitor contractor performance under this contract using the method of surveillance specified in the QASP and Performance Requirements Summary (PRS).

5.7.2 Performance measures will be determined for each individual requirement. Examples of specific performance measures include quality of product or service, schedule, cost control, customer satisfaction, safety, performance (including the number of CDRs), complexity of work, and management of key personnel.

5.8 System Security Engineering Maintenance and Support. The contractor shall adhere to Information Assurance regulations and directives and provide system security engineering. The contractor shall provide timely patches for vulnerabilities (vendor specific as well as globally published vulnerabilities), adhere to IAVAs, and maintain a favorable A&A decision and an ATC to all requisite networks for the life of the contract.

5.9 Key Personnel. The contractor shall notify the Government of any key personnel changes. The contractor shall not substitute any employee specifically named in its proposal without the prior written consent of the KO. The Government reserves the right to disapprove any changes that are not in the best interest of the Government. Key personnel shall, at a minimum, meet the requirements listed for each discipline shown in Appendix C.

5.10 Service Contractor Reporting. The Government operates and maintains a secure data collection site where the contractor will report ALL contractor labor hours (including sub-contractor labor hours) required for the performance of this contract. The contractor is required to completely fill in all the information in the format using the following web address <https://www.sam.gov> (contractor shall select the appropriate DoD service/component based on TO funding source). The required information includes:

- (1) Contracting Office, KO, COR.
- (2) Contract number, including task and delivery order number.
- (3) Beginning and ending dates covered by reporting period.
- (4) Contractor name, address, phone number, email address, identity of contractor employee entering data.
- (5) Estimated direct labor hours (including sub-Contractors).
- (6) Estimated direct labor dollars paid this reporting period (including subcontractors).
- (7) Total payments (including subcontractors).
- (8) Predominant Federal Service Code (FSC) reflecting services provided by contractor (and separate predominant FSC for each subcontractor if different).

- (9) Estimated data collection cost.
- (10) Organizational title associated with the Unit Identification Code (UIC) for the Army Requiring Activity (the Army Requiring Activity is responsible for providing the contractor with its UIC for the purposes of reporting this information).
- (11) Locations where contractor and subcontractors perform the work (specified by zip code in the United States and nearest city, country, when in an overseas location, using standardized nomenclature provided on website).
- (12) Presence of deployment of contingency contract language; and
- (13) Number of contractor and subcontractor employees deployed in theater this reporting period (by country).

As part of its submission, the contractor will also provide the estimated total cost (if any) incurred to comply with this reporting requirement. Reporting period will be the period of performance not to exceed twelve months ending 30 September of each year and shall be reported by 31 October of each year. Information from the secure web site is proprietary in nature when the contract number and contractor identity are associated with the direct labor hours and direct labor dollars. At no time will any data be released to the public with the contractor's name and contract number associated with the data. Reports and queries from the database shall not contain proprietary data.

5.11 MAINTENANCE AND SERVICE:

5.12 Preventive Maintenance: The Contractor shall perform all PM activities consistent with the operational requirements to ensure continuous operation and reliability of each system.

5.13 Corrective Maintenance: The contractor shall replace, or repair failed, damaged, or defective hardware, software, instrumentation, and controls on the equipment specified in the PWS. Repair may be overhaul, reprocessing, or replacement of nonfunctional parts or materials that have failed or deteriorated by action of the elements or usage and have not been corrected through maintenance, or it is the replacement of the entire unit or system if beyond economical repair. Repair work may also include inspection testing, cleaning lubrication, adjustment, calibration and part or component replacement and programming as required completing the service call repair.

5.14 Service Orders (SO): Service Orders will be used for CM services not covered in the PM portion of the contract as indicated in the individual requirement PWS. Ordering Officers (OO) may be appointed by the KO at the individual requirement level and are authorized to approve SOs at or below their designated threshold. Contractors will be provided a copy of the OO appointment letter as applicable. The contractor must ensure that they do not accept work above the OO's authority. All other SOs require approval by the KO. The contractor will normally prepare and submit a SO to the designated Government representative for routing and/or approval. The SO will describe the work to be performed, associated cost, and all required backup documentation. Upon approval, the contractor shall perform the work described on the SO.

5.15 Record Keeping: The contractor shall keep records/logs/checklists of all PM inspections that are performed that track the following at a minimum: date and time of inspection, asset name, asset tag information, locations, work performed, any parts replaced, and if there are any

corrective issues required from the inspection. The log shall be maintained for all required system devices, equipment, sub-systems, and components. Complete logs shall be kept and be available for inspection on site, demonstrating that planned and systematic adjustments and repairs have been accomplished for the equipment itemized in the individual requirement.

5.16 Specific task areas or functions that may be performed are listed below. All tasks are to be performed IAW the requirements of the specifications, drawings, original manufacturer's published procedures (if applicable), and CDRLs.

5.16.1 Perform site surveys, develop technical recommendations including assessment of existing equipment, identification and validation of problems and deficiencies, and provide repair/replacement/modernization recommendations.

5.16.2 The contractor must have sufficient spare parts or access to spare parts to meet delivery requirements.

5.16.3 Maintain, and update as necessary, accurate as-built drawings of the system.

5.16.4 Repair, maintain, and update systems, including associated data transmissions systems. Contractor shall comply with all requirements of the NEC, or its equivalent authority related to installation of new data transmission equipment and systems.

5.16.5 Provide, populate and maintain computerized maintenance management software system, including the generation of work tickets for maintenance activities.

5.16.6 Provide central monitoring services, including associated facilities, equipment, appurtenances, and trained monitoring personnel for the service of remotely monitoring alarm systems and taking actions as appropriate. This service may require a facility that is always staffed and in operation.

PART 6 APPLICABLE PUBLICATIONS

6.1 All technical and programmatic work performed under this contract and its TOs shall comply with the applicable criteria issued by DoD, Department of the Army (DA), and other Government Agencies within and outside DoD. The criteria documents issued by the Corps of Engineers shall be used as the primary criteria unless alternate, additional or supplementary criteria is stated in the individual TO. Applicable regulations issued by the DA with associated DoD issued policy and Department of Defense Instructions (DoDI), and/or Department of Defense Directives (DoDD) shall be followed. Where these regulations, policy and criteria are applicable to UMCS, ESS or related installation, testing, operation, management, etc., those requirements shall be incorporated. A list of applicable criteria is provided in contract document.

6.2 Individual requirements will include a list of regulations and criteria documents applicable for survey, analysis, design, and cost estimate preparation.

PART 7 SPECIAL REQUIREMENTS

7.1 In addition to complying with the requirements outlined in FAR Part 22.13, FAR 52.222-38, FAR 52.222-35, FAR 52.222-37, DFARS 222.13 and Department of Labor regulations, USACE contractors and subcontractors at all tiers are encouraged to promote the training and employment of U.S. veterans while performing under the resultant contract. USACE contractors are encouraged to seek out highly qualified veterans to perform services under this contract. The following resources are available to assist USACE contractors in their outreach efforts:

Federal Veteran employment information at <http://www.fedshirevets.gov/index.aspx>
Department of Labor Veterans Employment Assistance <http://benefits.va.gov/vow/>
Department of Veterans Affairs- VOW to Hire Heroes Act <http://benefits.va.gov/vow/>
Army Wounded Warrior Program- <http://wtc.army.mil/modules/employers/index.html>
U.S. Chamber of Commerce Foundation-Hiring Our Heroes <http://www.hiringourheroes.org/>
Guide to Hiring Veterans-Reference Material
http://www.whitehouse.gov/sites/default/files/docs/white_house_business_council_guide_to_hiring_veterans_0.pdf

PART 8 ANTITERRORISM (AT/OPSEC) REQUIREMENTS

8.1. General security requirements and guidance: The security requirements described below apply to all contract personnel (including employees of the prime Contractor (“Contractor”) and all subcontractor employees) supporting the performance requirements of this contract. The Contractor is responsible for compliance with these security requirements. Questions regarding security matters shall be addressed to the designated Government representative (e.g., Contracting Officer Representative (COR), Requiring Activity (RA) representative, or Contracting Officer (if a COR or other RA representative is not appointed)). Contract personnel are critical to the overall security and safety of US Army Corps of Engineers (USACE) installations, facilities and activities, and security awareness training contributes to those efforts. The Department of Defense (DoD) and Army security training requirements specified below, if applicable, are performance requirements; all applicable contract personnel shall complete initial training within 30 days of contract award or the date new contract personnel begin performance on the contract. Within five business days from the completion of training, the Contractor shall provide written documentation (e.g., email or memorandum) to the Government representative. The documentation shall include the names of contract personnel trained and which training they completed; the Contractor shall maintain training records as part of their contract files and be prepared to provide copies of training certificates to the Government representative. Contractor personnel and vehicles are subject to search when entering federal installations. Additionally, all contract personnel shall comply with Force Protection Condition (FPCON) measures, Random Antiterrorism Measures (commonly referred to as “RAMs”), and Health Protection Condition (HPCON) measures. The Contractor is responsible for meeting performance requirements during elevated FPCON and/or HPCON levels in accordance with applicable RA plans and procedures—this includes identifying mission essential and non-mission essential personnel. In addition to the changes otherwise authorized by the changes clause of this contract, should the FPCON or HPCON levels at any individual facility or installation change, the Government may implement security changes that affect contract personnel. The Contractor shall ensure all contract personnel are aware of their security responsibilities, including any site-specific requirements identified in local policies or procedures.

8.2. Antiterrorism (AT) Level I training: All contract personnel requiring routine access to Army installations, facilities, and controlled access areas, or requiring network access shall complete initial and annual refresher AT Level I awareness training. Online AT Level I awareness training is available at <https://jko.jten.mil/> (website subject to change).

8.3. Physical security and access control requirements: All contract personnel requiring physical access to a federal installation or facility shall comply with the access control procedures of that location. Contract personnel requiring unescorted access to meet contract performance requirements on a DoD installation in the US shall be vetted by the installation/facility Provost Marshal/Directorate of Emergency Services/Security Office using the National Crime Information Center-Interstate Identification Index (commonly referred to as "NCIC-III") and Terrorist Screening Database (commonly referred to as "TSDB"). Contract personnel shall comply with all personal identity verification requirements specified in installation/facility policies and procedures. Contract personnel who do not meet requirements for unescorted access to USACE facilities shall coordinate escorted access with the Government representative, as needed. Contract personnel who receive keys, access cards, or lock combinations that provide access to government-owned property shall comply with key and lock control procedures of the RA.

8.4. Contract personnel requiring a common access card (CAC): Contract personnel will be issued a common access card (CAC) only if duties involve one of the following: (1) both physical access to a DoD facility and access to DoD information systems or networks; (2) remote access to a DoD information system or network using DoD-approved remote access procedures; or (3) physical access to multiple DoD facilities or multiple non-DoD federally controlled facilities on behalf of the DoD on a recurring basis for a period of 6 months or more. Before CAC issuance, contract personnel must have, at a minimum, a favorably adjudicated Tier 1 investigation or an equivalent or higher investigation in accordance with applicable Army regulations and Homeland Security Presidential Directive-12 (HSPD-12). At the discretion of the RA, an initial CAC may be issued based on a favorable review of a fingerprint check and a successfully scheduled Tier 1 investigation with the National Background Investigations Bureau. The RA provides contract personnel with additional information and forms to initiate the CAC issuance process, and/or to initiate background investigations, when required. Contract personnel shall complete these processes within established timelines to avoid delays.

8.5. Security requirements for contract performance outside the US: For contract performance requirements that involve services or delivery in a foreign country, the Contractor shall comply with the requirements of DFARS clause 252.225-7043, as applicable. For performance requirements that involve contract personnel accompanying or supporting US Armed Forces deployed outside the US, the Contractor shall comply with the requirements of DFARS clause 252.225-7040, as applicable. Contract personnel accessing DoD or other federal facilities outside the US shall comply with applicable Status of Forces Agreements and Geographic Combatant Command requirements. Prior to contract personnel traveling outside the US, the Contractor shall provide documentation of AT, OPSEC, and other required training to the COR, or other RA representative. Additionally, contract personnel shall comply with travel requirements specified in the DoD Foreign Clearance Guide, including country and theater clearance processes.

8.6. Suspicious Activity Reporting training (e.g. iWATCH, CorpsWatch, or See Something, Say Something): All contract personnel shall receive initial and annual refresher training from the RA representative on the local suspicious activity reporting program. This locally developed training provides contract personnel with general information on suspicious behavior, and guidance on reporting suspicious activity to the project manager, security representative or law enforcement entity.

8.7. Contract personnel requiring access to Government information systems: All contract personnel with access to a government information system (including USACE business systems and CAC-enabled websites) shall comply with applicable DoD and Army regulations and shall use the organization's UserID-Password Administration and Security System (U-PASS) at commencement of services to request network user accounts. Contract personnel shall complete DoD Information Assurance Awareness training prior to accessing information systems, and annually thereafter.

8.8. Training requirements for the protection of sensitive information: All contract personnel with access to critical information (as identified in the RA's OPSEC Program) shall complete initial and annual refresher OPSEC Level I Awareness training, which is available at the following websites: <https://www.iad.gov/iass/>, or <http://www.cdse.edu/catalog/operations-security.html> (websites subject to change). All contract personnel with access to Controlled Unclassified Information (CUI) shall complete initial and annual refresher CUI training in accordance with applicable Army policy.

8.8.a. Cybersecurity Maturity Model Certification (CMMC) and CUI-FCI Compliance.

All contract personnel must comply with CMMC requirements, as detailed in the Defense Federal Acquisition Regulations Supplement (DFARS). To enhance the security of information and reduce risk of cyber-attacks the Department of War (DOW) established the CMMC Program, which requires pre-award assessment of information and covered contractor systems against established cybersecurity standards for safeguarding FCI and CUI.

8.9 Controlled Unclassified Information (CUI) or Federal Contact Information (FCI) will apply if the contact is available for public release. CUI or FCI will also apply if the contractor is required to have CUI residing in or transitioning through their information system at any time.

8.10 Cybersecurity Maturity Model Certification (CMMC).

CMMC is not required if CUI and FCI does not apply to this contract, only CMMC Level 1 applies.

CMMC Level 3 (self-certification + C3PAO + DIBCAC) applies if there is a Reasonable expectation that the release of the CUI would place National Security or Critical Systems (ex. Nuclear, cyber, missile defense)

CMMC Level 2 (self-certification) applies if there is not a Reasonable expectation that the release of the CUI would place National Security or Critical Systems (ex. Nuclear, cyber, missile defense)

CMMC Level 2 (self-certification + C3PAO) applies if any portion of the CUI anticipated in the contract and listed in the "Defense" Category

CMMC Level 2 (self-certification only) applies if no portion of the CUI anticipated in the contract and listed in the "Defense" Category

8.11. Information Assurance (IA)/Information Technology (IT) requirements: All contract personnel performing IA/IT services must comply with DoD training and certification requirements specified in DoD 8570.01-M, Information Assurance Workforce Improvement Program. Contract personnel shall provide the Government representative with documentation of requisite certification(s) prior to performing on the contract. In accordance with applicable DoD, Army, and USACE regulations, the Contractor shall ensure that all information systems (IS) and platform information technology (PIT) systems developed and/or supported under this contract comply with cybersecurity and architectural requirements, including, but not limited to: security

technical implementation guides (STIG)(e.g., the current version of the Application Security and Development STIG, and the internet access point (IAP) demilitarized zone (DMZ) STIG), and the use of security controls developed under the risk management framework documentation for the system or platform. The Contractor shall address questions regarding these provisions to the Government representative, who will coordinate between the Contractor and the USACE Chief Information Officer (CIO).

8.12. Escorting in classified and/or sensitive areas: In accordance with applicable regulations, all contract personnel who do not possess the appropriate security clearance or access privileges will be escorted in areas where they may be exposed to classified information or operations, sensitive information or activities, or restricted areas.

8.13. Pre-screen candidates using E-Verify Program: Contractors shall comply with the requirements set forth in FAR clause 52.222-54 Employment Eligibility Verification and FAR Subpart 22.18 in using the E-Verify Program at (<https://www.e-verify.gov/>) (website subject to change) to meet the contract employment eligibility requirements. Contractors are encouraged to cooperate with Federal and State agencies responsible for enforcing labor requirements to include eligibility for employment under United States immigration laws in accordance with FAR 22.102-1(i). An initial list of verified/eligible candidates shall be provided to the COR no later than three business days after the initial contract award. When contracts are with individuals, the individuals will be required to complete a Form I-9, Employment Eligibility Verification, and submit it to the Contracting Officer to become part of the official contract file.

APPENDIX A

1.0 Performance Requirements Summary (PRS): Contractor service requirements are summarized into performance objectives that relate directly to mission essential items. The performance threshold briefly describes the minimum acceptable levels of service required for each requirement. These thresholds are critical to mission success. All performance objectives listed in Table 2 may not apply to the specific project being performed. A project specific PRS will be included in the PWS at the individual requirement level.

Table 2

Procurement and Installation Performance Requirements Summary			
Performance Objective	PWS Ref Level (plus all other applicable areas)	Acceptable Performance Level (Standard)	Performance Threshold
1. Zero reportable safety incidents	Specified at TO level	IAW EM 385-1-1 and all applicable OSHA regulations	Zero reportable incidents
2. Complete Performance Verification Test (PVT)	Specified at TO level	IAW PWS requirements and applicable UFGS	First-time pass without significant punch-list items
3. Preparation and submittal of all applicable technical deliverables	Specified at TO level	IAW applicable UFGS, UFCs, and PWS requirements	On time delivery within Project Schedule and “A” or “B” grade in RMS
4. Submittal of security engineering and cybersecurity documentation	Specified at TO level	Meets requirements established in PWS	Minimal defects
5. Adherence to accepted project schedule	Specified at TO level	Within established contract period of performance	No more than 2 percent variance from critical path
6. Preparation and submittal of all non-technical deliverables to include contract compliance documentation	Specified at TO level	IAW contract deliverable table in PWS	Meet all contract reporting requirements

Maintenance and Service Performance Requirements Summary			
Performance Objective	PWS Ref Level (plus all other applicable areas)	Acceptable Performance Level (Standard)	Performance Threshold
1. Zero reportable safety incidents	Specified at TO level	IAW EM 385-1-1 and all applicable OSHA regulations	Zero reportable incidents
2. Quality and Timeliness of Deliverables	Specified at TO level	As outlined in the PWS.	90% compliance, Zero Deviation
3. Project Schedule	Specified at TO level	Within established period of performance	No more than 2 percent variance from critical path
4. Perform Preventive Maintenance	Specified at TO level	System is maintained IAW the PM schedule, and such that the systems are continuous and reliable.	95% Compliance
5. Perform Corrective Maintenance	Specified at TO level	System is repaired IAW PWS response times and procedures such that the systems are continuous and reliable.	95% Compliance

1.1 Types of Surveillance to select from:

1.1.1 Random Sampling: Appropriate for frequently recurring tasks. Evaluate randomly selected samples of the lot to determine the acceptability of the entire lot.

1.1.2 One-hundred Percent Inspection: Appropriate for tasks that occur infrequently. Inspect and evaluate performance each time task is performed

1.1.3 Periodic Surveillance: Evaluation of samples selected on other than 100 percent or statistically random basis per FAR 52.246-4, i.e. monthly, quarterly, semi-annually etc.

1.1.4 Customer Feedback: A method carried out by soliciting customer/end user satisfaction of technical services performed by the contractor.

1.2 Incentives: If contractor performance meets or exceeds contractual requirements.

1.2.1 Positive Past Performance evaluation rating within the CPARS.

1.2.2 Favorable consideration for continued service through contract option exercise (if applicable).

1.3 Consequences: Potential Government remedies for not meeting performance thresholds on a firm fixed price TO:

1.3.1 Government increasing the surveillance method that is stated in the QASP.

1.3.2 Possible Termination for Default.

1.3.3 Negative Past Performance evaluation rating within the CPARS.

1.3.4 Unfavorable consideration for continued service through not exercising the next contract option(s) (if applicable).

1.3.5 Government remedies identified in the services and/or construction clauses identified in the contract.

APPENDIX B
CYBERSECURITY ENGINEERING AND RISK MANAGEMENT FRAMEWORK (RMF)
COMMON TASK ORDER LANGUAGE AND PROCESS STEPS

A. CYBERSECURITY ENGINEERING TASK ORDER LANGUAGE (RMF Process Support NOT Required)

1.0 Cybersecurity Requirements: The contractor shall follow Unified Facility Criteria (UFC) 4-010-06 and Unified Facility Guide Specification (UFGS) 25 05 11, Cybersecurity of Facility-Related Control Systems.

UFC 4-010-06 defines the five steps to integrate cybersecurity into the FRCS Design as follows (see UFC 4-010-06 Chapter 3-1.1 Five Steps for Cybersecurity Design):

- a) Step 1: Based on the organizational mission and details of the control system, the System Owner (SO) and Authorizing Official (AO) determine the Confidentiality, Integrity, and Availability (CIA) impact levels (Low, Moderate, or High) for the control system. The CIA rating for the associated FRCS in this project is: [Designer / Project Engineer note: CIA rating must be obtained from stakeholder] Confidentiality: X (L,M, H) , Integrity: X (L,M,H), Availability: X (L,M,H)
- b) Step 2: Use the impact levels to select the proper list of controls from NIST SP 800-82. (SEE NOTE BELOW)
- c) Step 3: Using the DoD master Control Correlation Identifier (CCI) list, create a list of relevant CCIs based on the controls selected in Step 2. (SEE NOTE BELOW)
- d) Step 4: Categorize CCIs and identify CCIs that require input from the designer or are the designer's responsibility. (SEE NOTE BELOW)
- e) Step 5: Include cybersecurity requirements in the project specifications using the UFGS 25 05 11 and provide input to other specifications as required.

NOTE: Refer to Appendix H in the UFC which contains tables covering steps 2-4 as listed above. The contractor shall address the CCIs listed for the DESIGNER. The contractor shall provide design specifications to cover the requirement or provide input to others regarding the implementation or lack of feasibility of the CCI relating to the control system components installed, (workstations, servers, building point of connections, network switches, routers, and other IP enabled devices...). The DESIGNER controls will have an impact level of LOW or MODERATE. For LOW, LOW, LOW systems, address all DESIGNER controls listed as LOW. If the SO and AO determine the system is designated as MODERATE, the contractor shall address DESIGNER controls listed as LOW and MODERATE.

2.0 The Contractor shall assess (scan and perform manual checks) of the control system using DoD approved cybersecurity scanning tools (i.e. Nessus/SCAP) with current up to date definition files and applicable manual checks utilizing Defense Information Systems Assurance (DISA) Security Technical Implementation Guides (STIG) and Security Requirements Guides (SRG). All applicable STIGs and SRGs shall be applied (e.g. Windows server, SQL, Microsoft, Adobe, JAVA, or any application that has an applicable STIG or SRG). If vulnerabilities are found (Category (CAT) 1, 2, or 3) as the result of automated scans and manual checks, the Contractor shall remediate, mitigate, and or provide technical rationale as to why the vulnerability cannot meet compliance requirements. Upon completion of vulnerability resolution, the Contractor shall rescan the utilizing the same methods as previously stated to ensure all vulnerabilities have been properly remediated and / or properly and acceptably mitigated or documented as an open vulnerability. The Contractor shall strive to mitigate and or remediate all vulnerabilities associated with the control system to greatest extent possible. Any remaining CAT 1 and High-Risk CAT 2 vulnerability finding(s) that remain open findings are to be reported to CEHNC with supporting rationale as to why the finding(s) cannot meet compliance measures along with any mitigations and or compensating cybersecurity measures that lower the risk of the non-compliance.

3.0 Cybersecurity Deliverables are defined in each TO. Types of OS related deliverables are:

- a) All completed applicable STIG and SRG Checklist(s) to include Automated and Manual checks.
 - b) Proof of current patching/IAVA compliance of servers, workstations, network devices, etc... The contractor shall use current definition files for all scanning tools.
 - c) STIG / SRG Deviations list for any STIG / SRG requirement that cannot be applied.
 - d) Hardware & Software Inventory List
 - e) System / Network Topology Diagrams of the system depicting the Authorization Boundary
- Types of cybersecurity deliverables related to FRCS may include development of security guidance, checklists, and/or process documents. UFGS 25 05 11 further defines submittals to be delivered by the contractor when executing projects.

B. RISK MANAGEMENT FRAMEWORK SUPPORT TASK ORDER LANGUAGE:

1.0 Support execution of the Risk Management Framework (RMF) Process:

The Contractor shall provide the services to accomplish the tasks they have been awarded and as further described in the sections below as they relate to Cybersecurity. The tasks of this PWS may involve work related efforts such as but not limited to, interviewing the system owner and representative staff, facility security and management staff, system owners and end user operators, business process managers, and Cybersecurity support personnel. The tasks may also involve reviewing and updating existing documents and developing new documents such as but not limited to; system security plans (SSP), application system data flow diagrams,

comprehensive A&A packages, user guides, operational procedures, vulnerability management plans, risk assessments, and functional mission statements. At all times the contractor shall comply with existing DoD and USACE related authorities, policies and regulations, existing standards and guidelines from the National Institute of Standards and Technology (NIST), service / component derived and / or vendor specific best practices. Cybersecurity will be placed on the agenda of the post award orientation meeting for discussion and clarification upon Contractor's request. This will be achieved through abiding by all applicable Cybersecurity regulations and directives to ensure a favorable RMF A&A decision (i.e. ATO - Authority to Operate), as well as obtaining an Authority to Connect (ATC) to all requisite networks. All network infrastructure equipment (e.g. network switches, etc., NOT Building Level Controllers / Building Point of Connection devices) provided shall be listed on the Unified Capabilities Approved Products List (UC APL) and Trade Act Agreement (TAA) compliant. Unless the contracting officer provides written notification to the contrary, for the purposes of cybersecurity in this task order, the system includes the entire system (to include, but not limited to, ALL hardware, software and/or equipment), not just the parts of the system that the contractor has provided. The Contractor shall provide timely patches for vulnerabilities (vendor specific as well as globally published vulnerabilities), adhere to Information Assurance Vulnerability Alerts (IAVAs), adhere to Industrial Control Systems Cyber Emergency Response Team (ICS-CERT) alerts and notifications all servers/workstations and devices within the authorization boundary by the contractor up to the point of turnover to the Government. The Risk Management Framework (RMF) requires the Continuous Monitoring of implemented Cybersecurity controls and updates to the Enterprise Mission Assurance Support Service (eMASS) entry of all RMF documentation and artifacts to maintain a favorable A&A. Unless specified in a PWS/ TO, continuous monitoring is the responsibility of the System Owner post granting of the favorable A&A decision.

C. ASSESS AND AUTHORIZATION (RMF PROCESS)

1.0 BACKGROUND: DoDI 8500.01, Cybersecurity, and DoDI 8510.01, RMF for DoD IT, incorporates Platform IT (PIT) into the RMF process. PIT is a category of both IT hardware and software that is physically part of, dedicated to, or essential in real time to the mission performance of special purpose systems. The government will advise the contractor on the status and AO designation of the planned system as PIT. Categorization of the system will follow to identify the required controls IAW the RMF process.

1.1 The Contractor shall provide a cyber-secure system(s) with all applicable security artifacts and security engineering to meet the requirements of receiving an ATO accreditation decision via the DoD RMF. The implementation of cybersecurity measures in relation to design and construction / installation of the system shall impede the system's functional requirements. However, cybersecurity measures should be applied to the greatest extent possible and where compliance cannot be met, deviations from cybersecurity standards should be documented and appropriately justified. The expected duration for RMF Activities 1-5 stated below shall be approximately 12 months. The Contractor shall conduct and participate in RMF meetings as required by the PWS.

1.2 Implementation of RMF is a major component to attain secure control systems (CS). There are six (6) steps/activities to the RMF process, and the contractor shall assist and support the government as necessary and as defined within this PWS to ensure the completion of the steps listed below and as outlined in the following tasks:

- a) Categorize Information System
- b) Select Security Controls
- c) Implement Security Controls
- d) Assess Security Controls
- e) Authorize Information System
- f) Monitor Security Controls (Optional O&M Effort)

1.3 Guidance on RMF execution can be found at the following sites:

1.3.1 RMF Knowledge Service <https://rmfks.osd.mil/>

1.3.2 NETCOM RMF <https://army.deps.mil/NETCOM/sites/RMF/SitePages/Home.aspx> *

*Army Specific

1.4 The Contractor shall refer to the above sites for the most current guidance and information related to RMF execution. Access to the above sites require a valid CAC for authentication and access to the websites.

2.0 SUPPORTING CRITERIA: As they apply to the agency / service requirements of this specific task order, the Contractor shall provide Cybersecurity in accordance with current policies, procedures, and statutes, to include (but not restricted to) the following:

- a) 44 U.S.C. § 3542, January 2012.
- b) Committee on National Security Systems Instruction 6309, National Information Assurance Glossary, April 2010.
- c) Committee on National Security Systems Instruction 1253, Security Categorization and Control Selection for National Security Systems, March 27, 2014, as amended.
- d) Committee National Security Systems, Policy No. 11 National Policy Governing the Acquisition of Information Assurance (IA) and IA-Enabled Information Technology Products, June 2013, as amended.
- e) Federal Information Processing Standards Publication 199, Standards for Security Categorization of Federal Information and Information Systems, February 2004.
- f) Federal Information Processing Standards Publication 200, Minimum Security Requirements for Federal Information and Information Systems, March 2006
- g) Federal Information Security Management Act (P.L. 107-347, Title III), December 2002.
- h) Department of Defense Instruction 8500.01, Cybersecurity, dated 14 Mar 14.
- i) Department of Defense Instruction 8510.01, Risk Management Framework (RMF) for DoD Information Technology, dated 24 May 16.
- j) Department of Defense Directive 8140.01, Cyberspace Workforce Management, August 11, 2015.

- k) Department of Defense 8570.01-M, Information Assurance Workforce Improvement Program, Change 3 dated 10 Nov 2014.
- l) Defense Information Systems Agency (DISA) Secure Technical Implementation Guides (STIGs)
- m) Army Regulation 25-1, Army Information Technology, 25 June 2013.
- n) Army Regulation 25-2, Information Assurance, Rapid Action Revision Issue Date: 23 March 2009
- o) National Institute of Standards and Technology Special Publication 800-30, Guide for Conducting Risk Assessments, September 2012.
- p) National Institute of Standards and Technology Special Publication 800-37, Revision 1, Guide for Applying the Risk Management Framework to Federal Information Systems: A Security Life Cycle Approach, February 2010.
- q) National Institute of Standards and Technology Special Publication 800-39, Managing Information Security Risk: Organization, Mission, and Information System View, March 2011.
- r) National Institute of Standards and Technology Special Publication 800-53, Revision 4, Security and Privacy Controls for Federal Information Systems and Organizations, 22 Jan 2015.
- s) National Institute of Standards and Technology Special Publication 800-53A, Guide for Assessing the Security Controls in Federal Information Systems and Organizations: Building Effective Security Assessment Plans, 18 Dec 2014.
- t) National Institute of Standards and Technology Special Publication 800-59, Guideline for Identifying an Information System as a National Security System, August 2003.
- u) National Institute of Standards and Technology Special Publication 800-60, Revision 1, Volume I: Guide for Mapping Types of Information and Information Systems to Security Categories, August 2008.
- v) National Institute of Standards and Technology Special Publication 800-60, Revision 1, Volume II: Appendices
- w) Guide for Mapping Types of Information and Information Systems to Security Categories, August 2008
- x) National Institute of Standards and Technology Special Publication 800-82 Revision 2, Guide to Industrial Control Systems (ICS) Security, February 2015

2.1 Nothing in this document supersedes applicable laws and regulations unless a specific exemption has been obtained and can be provided in official memoranda.

3.0 RMF ACTIVITY 1- CATEGORIZE SYSTEM:

3.1 FIPS 199 / CNSSI 1253 Security Categorization:

The contractor shall in coordination with the prescribed System Owner and Authorizing Official (AO) support categorization IAW FIPS 199 and CNSSI 1253 of the system and document the results of the security categorization in the security plan. The contractor shall participate in categorization discussions with CEHNC, System Owner /User Representative (UR), and the Information System Security Manager (ISSM) as requested to support security categorization.

3.1.1 The Contractor shall ensure that a written subsection of the security plan covers FIPS 199 / CNSSI 1253 Security Categorization and Threat Assessment. The Contractor shall provide the

System Owner/CEHNC the Confidentiality, Integrity and Availability (CIA) concurrence memorandum for approval. The memo shall be uploaded into eMASS for AO concurrence.

3.1.2 All control systems overall CIA determination shall be considered low-low-low unless otherwise notified in writing by CEHNC.

3.1.3 The NIST SP 800-82 R2 overlay for ICSs shall be used as appropriate for the system. It is recommended that the current version of the Department of Homeland Security (DHS) Cyber Security Evaluation Tool (CSET) be used as a development tool for system-level documentary RMF artifacts.

3.2 Initiate the Security Plan:

The Contractor shall in collaboration with the System Owner initiate and develop the Security Plan. The Security Plan provides an overview of the security requirements, description of the CS (including system boundary) and describes the security controls in place or planned for meeting those requirements.

3.3 The Security Plan shall provide sufficient information to understand the intended or actual implementation of each security control employed within or inherited by the CS from a common control provider. The Security Plan shall also contain supporting appendices or as references to appropriate sources, other risk and security-related documents such as but not limited to a(n):

- a) Risk Assessment
- b) Privacy Impact Assessment
- c) System Interconnection Agreements (if applicable)
- d) Contingency Plan
- e) Security Configurations
- f) Configuration Management Plan
- g) Incident Response Plan
- h) Continuous Monitoring Strategy.

3.4 System Registration: Proper System Registration consists of (2) separate but co-dependent activities. First, the System Owner is responsible for registering the CS in their respective Component information system registry, and for the DA the system registry is the Army Portfolio Management System (APMS).

3.4.1 Upon proper registration in APMS the System Owner will receive an APMS ID, referred to herein as "DITPR ID". The DITPR ID will then support the second activity of System Registration, eMASS registration. The System Owner will be responsible for creating the initial eMASS registration of the system. The Contractor shall coordinate with CEHNC to obtain an eMASS account. The Contractor will be required to submit a completed DD Form 2875 and complete the eMASS Computer Based Training (CBT) (<https://disa.deps.mil/ext/cop/iase/epass/Pages/training.aspx>). Upon completion of the required training, CEHNC and the System Owner will ensure the eMASS administrators for the system apply the appropriate permissions for the contractor's access to the system registration. Access to eMASS will require the Contractor to have a valid CAC and access to the NIPRnet or SIPRnet as applicable. In some cases, not having access to a military (.mil) network can be mitigated by

using the DISA BootMe. The contractor will need to be provided a domain network account by the customer. Then, the customer and or USACE can request a custom DISA BootMe disk with the appropriate VPN setup into the customers .mil (dot.mil) network. This allows access too many .mil networks, including eMASS from any CAC enabled laptop or desktop.

3.4.2 The Contractor shall record the DITPR ID in the System Identification field within the Security Plan

3.5 Assigning RMF Roles: Identification and assignment of RMF Team members is a collaborative process involving all relevant stakeholders, to include the Contractor, which may interact with the system throughout its lifecycle.

3.6 The Contractor in coordination with CEHNC and System Owner shall ensure that the RMF Team positions and are properly identified and listed in the Security Plan. The RMF Team includes:

- a) Component Chief Information Officer (CIO)
- b) Component Senior Information Security Officer (SISO) (formerly the Senior Information Assurance Officer)
- c) Authorizing Official (AO) (formerly designated approving (or accrediting) authority)
- d) Authorizing Official Designated Representative (AODR) (if applicable)
- e) Security Control Assessor (SCA) (formerly certifying authority)
- f) Information System Owner (ISO)
- g) Program Manager or System Manager
- h) ISSM (formerly IA manager)
- i) Information System Security Officer (ISSO) (formerly IA officers)
- j) User Representative (UR)

3.7 The System Owner should also include representatives from the following roles to participate in RMF activities as well to ensure that the CS functional level areas of expertise are accounted for in the development of the RMF strategy and initial definition of the security boundary:

- a) Facility Engineer/Manager
- b) Facility Operations & Maintenance/Technician
- c) Physical Security Specialist
- d) Emergency Manager
- e) Organizational IT Network/Communications Specialist
- f) Organizational Information Assurance Specialist
- g) Installation Tenants (as required)
- h) Installation Network / C4IM Service Provider ISSM
- i) Operations and Maintenance Contractors
- j) Control System Vendor/Integrators
- k) Information Assurance Contractor

3.8 Upon completion of RMF Activity 1- Categorize System, the Contractor shall, provide the following system-level documentary RMF artifacts as deliverables:

- a) C-I-A AO Concurrence Memorandum
- b) System Architecture / Network Topology / Data Flow depicting CS Authorization Boundary
- c) Hardware Inventory List
- d) Software Inventory List
- e) Ports, Protocols, and Services (PPS) List in accordance with DoD PPS Category Assurance List (CAL)

4.0 RMF Activity 2- Select Security Controls:

4.1 RMF Control Selection:

The Contractor in coordination with the RMF Team shall select the Security control baseline* (based upon the System Categorization completed in RMF Activity 1) , Select and apply security control overlays (for CS the NIST 800-82 R2 ICS Overlay*), Conduct appropriate tailoring of the selected set of security controls to include; (Supplementing the tailored baseline security control set, and identification of Inherited Common / Hybrid Security Controls), and documenting the Final Security Control Set to be approved in the Security Plan.

4.2 Select NIST SP 800-53 Rev 4 (note: NIST SP 800-82 Rev 2 Industrial Control Systems Security Guide, Appendix G ICS Overlay will be used to evaluate control systems and will eventually be programmed into eMASS. Until that happens, the EI&E NIST SP 800-82 Rev 2 ICS Overlay Security Controls and EI&E NIST SP 800-53 Revs 4 and 800-82 Rev 2 Merged found at (<https://rmfks.osd.mil/>) shall be used manually to complete the security controls until the eMASS functionality is available.

4.3 The Contractor shall ensure that the selected set of security controls, along with the supporting rationale for selection decisions and any system use restrictions, are included in the security plan. The security plan must also, identify all common controls inherited from external providers, and establish minimum assurance requirements for those controls.

4.4 RMF Team Members directly and indirectly involved in the security control selection are:

- a) Authorizing Official (AO)
- b) ISSM or ISSO if assigned
- c) ISO /Program Manager
- d) Information Owner (IO)
- e) Operations and Maintenance Contractors
- f) Control System Vendor/Integrators
- g) Information Assurance Contractor

4.5 Continuous Monitoring Strategy:

In coordination with the System Owner and the local network service provider (NEC, Army Corps of Engineers IT (ACE-IT), or other Component / Service Branch (USN, USMC, AF, DLA, etc., Network Service Provider), if applicable, the Contractor shall develop a system-level strategy for the continuous monitoring of the CS. The system-level continuous monitoring strategy is a mandatory requirement to be included in the Security Plan.

4.6 If the CS is a closed restricted network or standalone, it will be the responsibility of the System Owner to meet the requirement for continuous monitoring through their own respective means. The Contractor with coordination through the System Owner shall develop a system-level strategy for the continuous monitoring of the CS. The system-level continuous monitoring strategy is a mandatory requirement to be included in the Security Plan.

4.7 Approval of the Security Plan:

Upon completion of security control selection within eMASS, the Security Plan can be generated. The Contractor shall submit through CEHNC to the System Owner the completed Security Plan for review and approval. Upon approval by CEHNC and the System Owner the Security Plan to include the system-level continuous monitoring strategy will be submitted to the AO (or designee) by the System Owner. By approving the security plan, the AO agrees to the system categorization, the set of security controls proposed to meet the security requirements for the system, and the adequacy of the system-level continuous monitoring strategy.

4.8 Upon Contractor completion of RMF Activity 2- Categorize System, the Contractor shall, provide the following system-level documentary RMF artifacts as deliverables:

- a) Initial Security Plan
- b) Continuous Monitoring Strategy
- c) Risk Assessment
- d) Privacy Impact Assessment
- e) System Interconnection Agreements (if applicable)

5.0 RMF ACTIVITY 3- IMPLEMENT SECURITY CONTROLS:

5.1 Internal Security Control Self-Assessment Preparation: The Contractor shall develop, review, and obtain CEHNC and System Owner approval of a plan to internally assess the selected system-level security controls. The plan should include identification of assessment testing logistics, vulnerability scanning tools (i.e., ACAS/Nessus, SCAP, SRG, etc.) roles and responsibilities, detailed test plans, etc.

5.2 Internal Security Control Self-Assessment (Scan/Fix/Scan Testing and Analysis): The contractor shall assess (scan and perform manual checks) of the control system using approved cybersecurity scanning tools and applicable manual checks utilizing Defense Information Systems Assurance (DISA) Security Technical Implementation Guides (STIG) as described within the Internal Security Control Self-Assessment Plan. If vulnerabilities are found (Category (CAT) 1, 2, or 3) as the result of automated scans and manual checks, the Contractor shall remediate, mitigate, and or provide technical rationale as to why the vulnerability cannot meet compliance requirements. Upon completion vulnerability resolution, the Contractor shall rescan the CS utilizing the same methods as previously stated to ensure all vulnerabilities have been properly remediated and / or properly and acceptably mitigated or documented as an open vulnerability. The Contractor shall strive to mitigate and or remediate all vulnerabilities associated with the CS to greatest extent possible. CAT 1 and High-Risk CAT 2 vulnerability

findings that cannot be remediated or mitigated to a lower acceptable risk as determined by the AO, are to be reported to the CEHNC with supporting rationale, and properly identified on the Initial Plan of Actions and Milestones (POA&M). The results of the Internal Security Control Self-Assessment shall be documented within eMASS by the System Owner with support from the Contractor.

5.3 The Contractor shall support the System Owner as necessary in associating a system-level artifact (in eMASS) with each Control Correlation Identifier (CCI) when the artifact is used to support test results from the Internal Security Control Self-Assessment. The contractor shall use the artifact name in the test results for example: “Documented in the SystemName System Security Plan...” vice “Documented in the SSP...”

5.4 The Contractor shall identify the specific section (using document section name) that satisfies each actionable item for each CCI. For example, “...in the SystemName System Security Plan, section/paragraph (Account Control, Purpose & Scope)...” vice “...in the SSP...”

5.5 The Contractor shall address each actionable item in each CCI for the Selected Security Controls determined by the RMF Team in RMF Activity 2- Select Security Controls. Internal Security Control Self-Assessment in eMASS should occur no later than 5 months prior to the Operational Need Date (OND) of the CS.

5.6 Upon completion of the Internal Security Control Self-Assessment the Contractor shall assist CEHNC and the System Owner to support the scheduling of the Security Control Assessor-Validator (SCA-V) (formerly referred to as the Agent for the Certification Authority (ACA)) utilizing the NETCOM Bid Tracker to solicit the services of a SCA-V Team (Army RMF efforts only). CEHNC and the System Owner may choose to reach out directly to any of the approved SCA-V Teams without utilizing Bid Tracker. The scheduling of the SCA-V shall need to occur no later than 6 to 8 months from OND. The Contractor shall account for this milestone event in their overall project schedule.

5.7 The selection of the SCA-V / ACA for other Component / Service Branches will follow their respective prescriptive process.

5.8 Upon Contractor completion of RMF Activity 3- Implement Security Controls, the Contractor shall, provide the following system-level documentary RMF artifacts as deliverables as required:

- a) Configuration Management Plan
- b) Disaster Recovery Plan
- c) Continuity of Operations / IT Contingency Plan
- d) Incident Response Plan
- e) Information Assurance Vulnerability Management (IAVM) / Patch Management Plan
- f) Physical Security Plan
- g) System Configuration Guide
- h) System Restoration Checklist
- i) Plan of Action and Milestones (POA&M)

- j) Updated Security Plan to reflect Internal Security Control Self-Assessment.
- k) Continuous Monitoring Plan

5.9 It is recommended that the current version of the DHS CSET or the PEO STRI Keystone RMF Workflow Enhancement Tool be used as a development tool for system-level documentary RMF artifacts. Request a Keystone account by clicking the “Register” hyperlink at <https://www.lt2portal.mil/>. Additional access and user information for the Keystone tool shall be provided after award as needed.

6.0 RMF ACTIVITY 4- ASSESS SECURITY CONTROLS:

6.1 SCA-V RMF Assessment: The Contractor shall be available as needed to assist the SCA-V team for the duration of the assessment. The normal duration of an assessment is typically 3-5 working days (on-site) for the scans, document review, and manual checks. It typically takes an additional 30 days (off-site) for the SCA-V team to perform analysis and provide the requisite reports. The contractor shall plan its schedule accordingly to effectively capture these events and the associated timelines to complete.

6.2 Consideration: The Contracting Officer may require consideration from the Contractor in the form of a credit to the contract price, equivalent to reimbursing the cost of the SCA-V assessment (including, but not limited to the labor and travel of personnel conducting the reassessment) if the Contractor’s efforts in providing RMF services does not meet acceptable passing criteria (ATO) of the initial SCA-V assessment or any other part of the A&A process.

6.3 Security Assessment Results & Remediation Actions:

The Contractor shall upon receipt of the Security Assessment Results conduct remediation actions based on the findings and recommendations of the SCA-V. Receipt of the SCA-V Final Security Control Assessment Results should be delivered to the System Owner no later than 30 days after RMF assessment.

7.0 RMF Activity 5- Authorize Information System:

7.1 POA&M: The Contractor shall prepare the POA&M based on the findings and recommendations of the Security Control Assessment Results and submit to CEHNC and the System Owner for review and approval. The Contractor shall create a vulnerability and one milestone for every N/C Control in the POA&M module of eMASS for Security Control Assessor-Representative (SCA-R) use in performing final residual risk analysis. The Contractor shall allot 15 days for POA&M development and 30 days for CEHNC and System Owner review and approval.

7.2 Security Authorization Package: Upon completion of CEHNC and System Owner approval of the POA&M the Contractor shall assist the System Owner in submission of the Security Authorization Package via eMASS into the Package Approval Chain (PAC) for review and approval. The PAC is tiered approval process in which the Security Authorization Package goes through multiple levels of approval by the PAC. The PAC consists of the following eMASS roles:

- a) System Owner
- b) Organizational ISSM (if applicable)
- c) Program ISSM
- d) Security Control Assessor-Representative (SCA-R)
- e) Security Control Assessor-Army (SCA-A)
- f) AO

7.3 If during the PAC Initial Review, the SCA-R identifies there is insufficient information or missing data or artifacts, the package must be returned to the Program ISSM for rework. The Contractor shall support the System Owner in resolving any issues related to rework within scope as directed by the COR. The Contractor shall allot for 25 days for the PAC review prior to submission to the AO for the final Security Authorization Determination. Security authorization package deliverables:

- a) System Security plan
- b) Security Control Assessment Report
- c) Plan of Action and Milestones
- d) Continuous Monitoring Plan

7.4 Security Authorization Determination: The AO makes the final approval decision in 6th level of the PAC, the Contractor shall allot 10 business days for the final AO Security Authorization Determination i.e., ATO

7.5 ATC: Upon receipt of the final Security Authorization Determination, the Contractor shall assist the System Owner in obtaining ATC via network service provider (NEC, ACE-IT, or other Component / Service Branch (USN, USMC, AF, DLA, etc. Network Service Provider), if applicable. The Contractor shall follow the network service provider's criteria and process for ATC. 60 days should be allotted for completion.

7.6 RMF Activity 6- Monitor Security Controls:
(Optional O&M Effort)

7.6.1 RMF Activity 6- Monitor Security Controls, is solely a System Owner responsibility. However, the System Owner can augment this requirement via an O&M Services contract to provide some of the requirements of long-term continuous monitoring which is in line with the Continuous Monitoring Strategy developed in RMF Activity 2- Select Security Controls.

APPENDIX C

QUALIFICATION OF CONTRACTOR PERSONNEL

Program Manager

The PgM shall have full authority to act for the contractor on all contract matters relating to daily operation of this contract. Typical seven to ten years relevant experience in the UMCS/ESS/FAS/SCADA technology field, including five years of experience in a management or supervisory capacity and five years in the functional area of the program to be managed. PgM proposed shall have a bachelor's degree in engineering or business from an accredited university or college, or a related field, or in the project's functional area.

Project Manager

Responsible for planning and executing major portions of the project. Prepares and maintains the project scope, schedule, and budget. Primary point of contact for the customer on assigned tasks. Oversees all work and takes corrective action as necessary to ensure success of assigned tasks. Manages staffing, budget, prioritization, and other personnel matters. May serve as technical lead for assigned tasks. Experience with Government or industry processes, procedures, standards, methodologies, or tools as relative to the assignment. Five years' experience in the project management field plus demonstrated experience in the project's functional area. Bachelor's degree from an accredited university or college. PMP certification preferred.

Contracts Manager

Shall have the full authority as a Company representative in all matters relating to contracts to include but not limited to contracts negotiations, modifications, and proposal submittals. The person proposed shall have ten years' experience in the contracting field, including five years of experience in a management level capacity. The person proposed shall have a bachelor's degree in business from an accredited university or college, or a related field, or in the project's functional area.

Quality Control Manager

Typical seven to ten years relevant experience in the quality control field, including five years' experience as a Quality Control Manager. The person proposed shall have a bachelor's degree from an accredited university or college, in a related field, or in the project's functional area.

Fire Protection Engineer

Will represent the company with fire alarm system engineering products and solutions and the adherence to applicable codes. Provides Professional Engineering Stamp for fire alarm systems. The person proposed shall have ten years' experience in the Fire Protection engineering field, including five years in the functional area of the project to be managed. Must have a U.S. Professional Engineers certification. The person proposed shall have a bachelor's degree in engineering from an accredited university or college, or a related field, or in the project's functional area.

Site Superintendent

Provides site wide management of personnel for routine activities according to established procedures or instructions. Complies with the standards and organizational requirements relative to specific assignments. Five years of experience related to specific heating ventilation and air conditioning experience with associated controls. Experience with Government or industry processes, procedures, standards, methodologies, or tools as relative to the assignment. High School diploma, plus technical training in field related to task assignment.

HVAC Technician

Provides support for HVAC/Controls activities according to established procedures or instructions. Works independently to assist with technical support for elements such as: Complex processes, Structural elements, Electric/electrical/electronic components, Equipment, Applications, Systems, Software, Networks, Satellites, Telecommunications, Facilities, or Machinery. Complies with the standards and organizational requirements relative to specific assignments. Five years of experience related to specific heating ventilation and air conditioning experience with associated controls. Experience with Government or industry processes, procedures, standards, methodologies, or tools as relative to the assignment. High School diploma, plus technical training in field related to task assignment.

Senior Security System Engineer (SSE)

The Senior SSE is responsible for the design, development, implementation, and/or integration of a DoD IA architecture, system, or system component for use within CE (computing environment), NE (network environment), and/or enclave environments. Must meet the DoDD 8570.01 "Information Assurance Training, Certification, and Workforce Management" and DoD 8570-M "Information Assurance Workforce Improvement Program" requirements for Information Assurance Manager (IAM) Level 3 and Information Assurance Technical (IAT) Level 3, or Information Assurance System Architect and Engineer (IASAE) Level 3.

Security System Engineer (SSE)

The SSE is responsible for the design, development, implementation, and/or integration of a DOD IA architecture, system, or system component for use within CE (computing environment), NE (network environment), and/or enclave environments in support of higher-ranking IA professionals. Must meet the DoDD 8570.01 "Information Assurance Training, Certification, and Workforce Management" and DoD 8570-M "Information Assurance Workforce Improvement Program" requirements for IAM Level 1 or 2, and IAT Level 1 or 2, OR IASAE Level 1 or 2.

Senior Electrical/Mechanical Engineer

Will represent the company with engineering products and solutions. The person proposed shall have ten years' experience in the Electrical/Mechanical engineering field, including five years in the functional area of the project to be managed. The person proposed shall have a bachelor's degree in engineering from an accredited university or college. Must have a Professional Engineers (PE) certification.

Electrical/Mechanical Engineer

Will provide mechanical/electrical engineering products and solutions. Five years' experience in the Electrical/Mechanical engineering field, including three years demonstrated experience in the functional area. Bachelor's degree in engineering from an accredited university or college.

Senior Electronic Security Engineer

Will represent the company with ESS system engineering products and solutions. The person proposed shall have ten years' experience in the electronic systems/information technology field and demonstrated experience in the project's functional area. The person proposed shall have a bachelor's degree in engineering from an accredited university or college, or a related field, or in the project's functional area. Must have a PE certification.

Electronic Security Engineer

Will represent the company with ESS engineering products and solutions. The person proposed shall have five years' experience in the electronic systems/information technology field and demonstrated experience in the project's functional area. The person proposed shall have a bachelor's degree in engineering from an accredited university or college, or a related field, or in the project's functional area.

Network Specialist

Responsible for the design, implementation, and defense of the network. Must meet the DoDD 8570.01 "Information Assurance Training, Certification, and Workforce Management" and DoD 8570-M "Information Assurance Workforce Improvement Program" requirements for IAT (Information Assurance Technical) Level 1 or 2, OR IASAE (Information Assurance System Architect and Engineer) Level 1 or 2 OR meet the requirements for any type of Computer Network Defense (CND) professional (CND-A, CND-IS, CND-IR, CND-AU) mentioned in the aforementioned regulations.